

A New Approach To Evaluate Layer Productivity Before Well Completion

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Summary

This work presents a new methodology to identify fluid in formation layers and estimate individual reservoir parameters using openhole formation testing techniques. Real-time pressure and fluid identification data are obtained from a new wireline formation testing tool. The tool's dual packer module is needed to isolate individual zones. Field cases from wells located in the San Jorge Gulf basin and in the Neuquen basin illustrate the validity of testing methodology. The reservoir permeability (vertical and horizontal) and formation damage are calculated from pressure transient analysis of buildup and interference pressure data taken from several wells. The results obtained using this technique is consistent with those obtained from testing after well completion (cased hole drillstem test). Evaluation results can be used to decide whether to complete or abandon the tested zone. Fluid type is identified in real time using an optical fluid analyzer. Evaluation of anisotropy on a productive zone scale from the vertical interference test is presented.

Introduction

The main objective of formation evaluation at openhole conditions is the identification and description of hydrocarbon reserves with the best degree of resolution possible to assist in deciding whether to abandon or complete the tested interval. Equally important is obtaining the reservoir pressure and reservoir parameters to compute formation permeability and transmissibility. These are usually measured using well logging and testing techniques both at open- and cased-hole conditions. However, success in identifying the correct fluid rarely exceeds 60% for reservoirs such as the one in the San Jorge Gulf basin, located in the central Patagonia region in southern Argentina, although in many cases a complete set of logs is used. This leads to completing, perforating, and testing all prospective intervals, which has proven to be an expensive evaluation and completion process.

The main reason for the unpredictable results is the multilayer nature of the gross prospective producing interval. The interval thickness of interest in a typical well is between 800 and 1200 m, with approximately 40 lenticular reservoirs ranging from 1 to 10 m thick. As shown in the San Jorge Gulf basin stratigraphic sequence in Fig. 1, the upper intervals are laminated sands with a high clay content. The bottom layers are tuffaceous sands of a variable, complex lithology. The sands are highly laminated, with a variable, high water saturation. In addition, the layers are laterally discontinuous (1 to 3 km wide) and heterogeneous. The initial oil production rate is about 30 m³/d (usually obtained by fracturing) and most of the wells are produced by rod pumping.

The major challenges in the San Jorge Gulf basin for the past 60 years have been to identify the potential oil layers in a multilayer system and to determine the expected production rate, reservoir permeability, and formation damage (mainly for fracture

design) for each potential layer prior to completing the well or zone. Early reservoir evaluation is necessary to provide these answers and because wells are put on rod pumping, which limits the subsequent use of direct evaluation methods.

Our research over the past 2 years in formation evaluation and testing techniques has focused on determining the applicability of new methods that may optimize current evaluation practices. As a result, a methodology based on application of the new-generation wireline modular formation dynamics tester (MDT) tool for evaluating layer productivity before well completion was implemented. In this paper, we present several field cases showing the independent evaluation of a given layer in a multilayer system. In these examples, formation fluid identification (besides mud and filtrate) is accomplished in real time, which assists in pressure-volume-temperature (PVT) analysis. Layer anisotropy is obtained and validated for the layer thickness scale. The values of permeability and formation damage are determined from pressure transient analysis of drawdown and buildup data obtained by isolating the layer using the tool's dual packer module. Testing time per layer is less than 1 h. Evaluation is done in real time, hence, the decision whether to abandon or complete the particular layer can be made at the wellsite. Even though the methodology presented here was applied mainly to the San Jorge Gulf Basin reservoirs, it is not limited to this basin, but it is valid for any laminated, multilayer, thick reservoir. A field case of a well completed in multilayer reservoir located in the Neuquen basin, Neuquen province, Argentina is presented to illustrate the validity of the method in a completely different formation geological environment.

Formation Evaluation and Testing

A typical formation evaluation and testing program in the San Jorge Gulf basin consists of running an appropriate suite of logs (usually including a conventional formation tester) to identify the layer's reserves. Openhole drillstem tests (DSTs) are also run as necessary. Completion of the well is based on the evaluation results. Casing is set, and all the prospective intervals are shot separately. Following a pressure buildup test of about 8 to 10 h duration [(TST) test], swabbing is conducted for 6 to 8 h in selected layers. The main objectives are to obtain fluid type, production rates, reservoir permeability, and formation damage (skin). These parameters are important for fracture design and equally important to define candidate zones for fracture treatments. Layer point pressures obtained with the conventional formation tester are an important measurement but limited in the case of a heterogeneous layer, such as a naturally fractured layer. The number of layers present in a well (up to 40) limits an exhaustive evaluation for economic reasons. For example, it would require a long time (days) to perform a cased hole DST for each layer present in a well. A brief summary of formation and testing evaluation limitations, based on experience, is as follows.

- The heterogeneous nature of the lithology usually requires conducting conventional testing on a very small vertical scale (centimeters), which limits the application of openhole DSTs.
- Verification of fluid identification using log techniques is usually done after a well or particular interval is completed, that is, cemented and perforated.

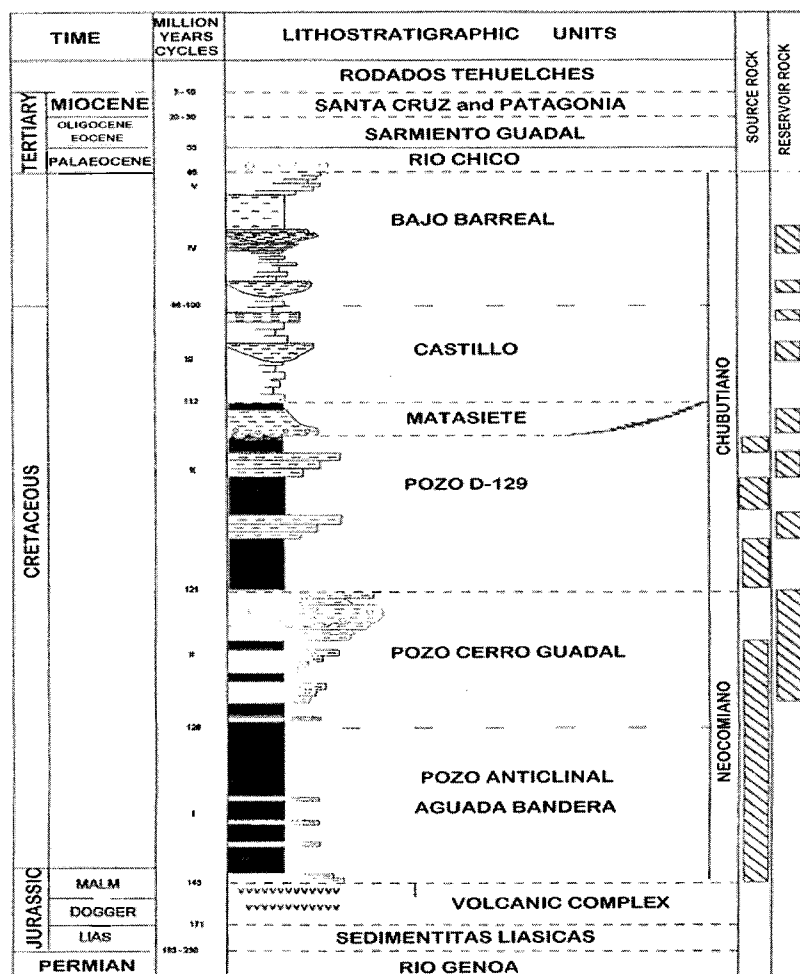


Fig. 1—San Jorge Gulf basin lithostratigraphic units. Formation evaluation and testing were conducted mainly in layers of the Chubutiano.

- The highly laminated nature of some layers requires an estimate of the layer anisotropy (which requires the vertical transmissibility parameter), which cannot be obtained with current testing techniques.

- Use of conventional formation testers to determine pressure in a layer and fluid sampling are limited in the case of naturally fractured layers. Samples are usually not representative of formation fluids other than mud and/or filtrate.

- Layer reservoir parameters such as permeability and formation damage, which are obtained from pressure transient analysis of buildup tests, are usually unreliable. This is because the flow and pressure history preceding the buildup period is usually not well defined in duration, associated rate, and corresponding pressure values (swabbing periods). Therefore, the validity of the obtained parameters is subject to many assumptions.

The aforementioned formation evaluation and testing limitations and the production of wells by rod pumping drove the search for new evaluation methods applicable at openhole conditions. Layer evaluation—in particular, fluid content and potential—should be done at the earliest stage possible, that is, at openhole conditions.

New Approach to Formation Testing

A detailed description of the MDT tool applied as part of the logging program for wells in the San Jorge Gulf basin or the Neuquen basin is presented in Ref. 1. This new formation tester evolved from conventional formation testers to address the previous gap from their limitations on identifying virgin formation

fluid and determining permeability and damage. In addition, the MDT tool enables obtaining layer anisotropy to enhance layer evaluation and description.

The formation and testing evaluation results obtained from several wells with the MDT tool were used to develop a new approach to enhanced layer productivity evaluation in the San Jorge Gulf basin. The basic idea is to combine geology and reservoir

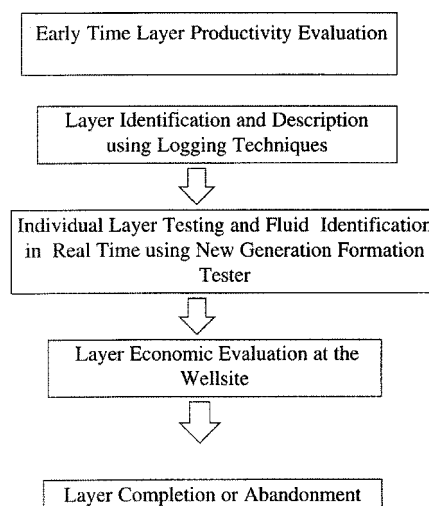


Fig. 2—Basic sequence of steps for early-time layer potential evaluation process in the San Jorge Gulf or any multilayer basin.

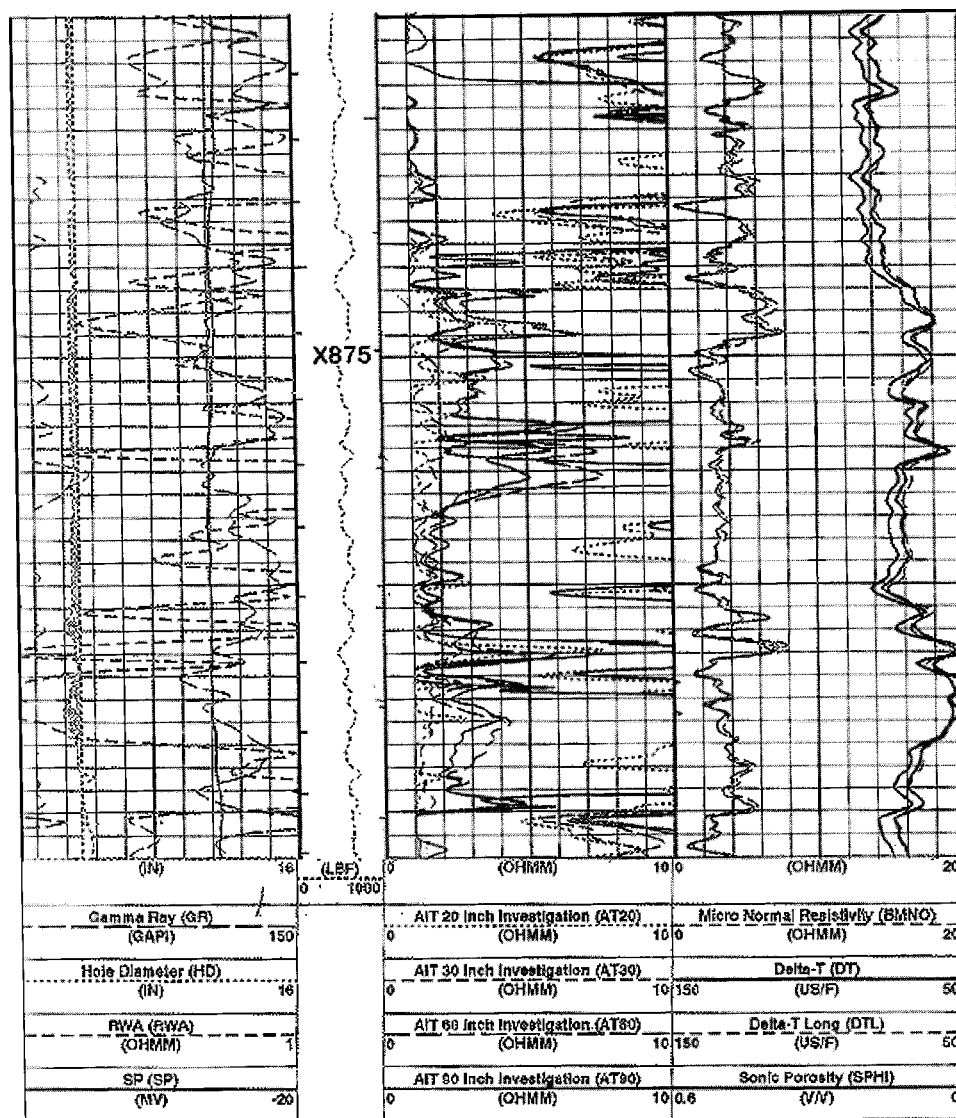


Fig. 3—Induction logs showing the layer resistivity response for the interval under study in field case 1.

engineering evaluation technologies at the earliest possible time of the “life” of the layer. The resulting logging program should provide data that allow determining petrophysical properties with good vertical resolution in order to detect pay zones.

As part of the logging program, the formation tester is set at the selected layer and the type of fluid predicted from logging data is then confirmed at the wellsite in real time. For low-permeability layers, the tool is made up with a dual packer module that allows better definition for dry tests. The dual packer module exposes a 1 m interval from a layer for evaluation. This configuration is also used to conduct layer testing in real time: a sequence of drawdown and buildup events can be conducted at the layer of interest. Rates as high as 4 m³/d were reported in oil zones. Conventional pressure transient analysis is applied to the obtained high-resolution pressure and rate data to estimate permeability and formation damage at the wellsite with a high degree of confidence.

Besides confirming logging predictions regarding the expected fluid type, the degree of anisotropy can be estimated (k_v/k_h). This is accomplished by conducting a vertical interference test using sink and monitor probes (minimum separation of 70 cm). The anisotropy value obtained at the wellsite can be used to guide perforation strategy.

Total testing time per layer depends on transmissibility, however, in most cases the total testing time was about 1 h. This procedure can be applied to all layers of interest without having to

pull the tool out of the hole. Fig. 2 shows the basic sequence of steps involved in this new approach for early-time evaluation of layer potential.

Field Applications

This new approach for layer evaluation has been applied successfully in many wells in the San Jorge Gulf basin. Three cases are presented here to illustrate the method. In addition, a field case of a well completed in the Neuquen basin is presented to show the application of the method in a different basin.

Field Case 1. The main objective of evaluation in this well was to confirm the presence of natural fractures and laminations. A secondary objective was to obtain an estimate of the formation pressure using a conventional formation tester at four points in the interval between X864 and X900 m. An openhole DST was conducted by isolating the entire interval (Fig. 3 shows resistivity logs for the interval studied). The results of the DST confirmed the presence of oil.

In an attempt to refine the vertical resolution and because the conventional formation tester found the four tested points dry (no entry), the MDT tool with the dual packer module was employed. The tool was set at X874.1 m. Once the 1 m interval was isolated in the layer by inflation of the packers to establish a seal against the hydrostatic pressure, drawdown was induced using the tool's

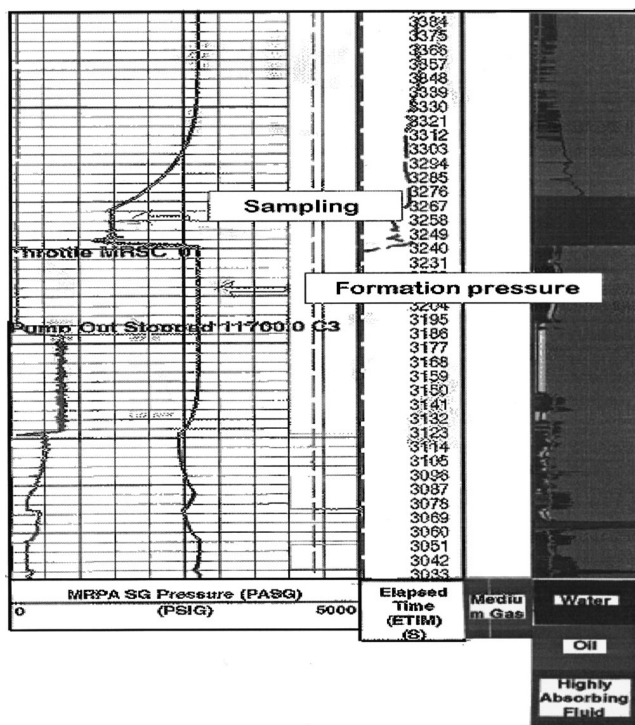


Fig. 4—Real-time fluid identification in field case 1 using the results from the OFA analyzer and wellbore pressure-transient behavior during flow, shut-in, and sampling periods.

pumping system. Real-time identification of the fluids entering from the formation was monitored using the optical fluid analyzer (OFA) module, which enables identifying the presence of oil, water, or gas in the line. Fig. 4 shows the fluid identification and corresponding transient pressure. The flowing pressure is not stable. The sequence of events is characterized by intermittent flow (about 1.9 m³/d) and shut-in. This effect is probably due to the high viscosity of the mud used. At about 2,655 s (0.7 h) total testing time (not shown on the figure because of space limitations) including packer inflation, oil entered the flowline. At this moment, testing of the layer was attempted to obtain the values for permeability and damage. Finally, a sample was taken to verify the surface prediction of fluid type. The fluid contents of the sample chamber were as expected: oil, mud, and filtrate.

Conventional pressure transient analysis was performed on the high-resolution data obtained using the pressure gauge of the packer module [Ref. 2]. Figs. 5–8 show diagnostic plots corresponding to the buildup period prior to sampling (0.073 h dura-

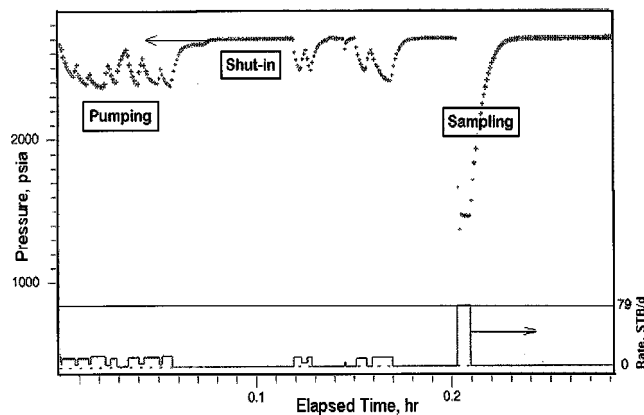


Fig. 5—Sequence of events (field case 1). During drawdown periods the flow rate was not constant because of the high mud viscosity.

tion) and test validation as a quality check on the calculated values of permeability and damage ($k=116$ md and $S=67$). The estimated formation pressure was 2,697 psi.

The diagnostic plot of the pressure and pressure derivative shows that a true radial flow regime was not reached during the buildup time (in part because of the high wellbore storage and skin effects). In this case, in addition to type curve and semilog analysis, nonlinear regression techniques were used to enhance interpretation of the data.

The parameters obtained should be considered valid because of validation of the testing flow rate history (see Fig. 8).

The well was cased, and a TST was conducted in the same interval. Conducting this test requires swabbing the well for about 8 h (no pressure measurements were taken during this period) and finally shutting in the well. The flowing pressure is obtained by letting fluid enter the well (increasing the liquid level) and measuring the pressure during the period (1 to 2 h) preceding the buildup (8 to 10 h). Figs. 9 and 10 show diagnostic plots of the pressure and pressure derivative for the buildup period.

Pressure transient analysis of buildup requires many assumptions, in particular, assuming that the wellbore flowing pressure is not totally affected by wellbore storage effects during the swabbing period. Validation cannot be performed for quality control of the analysis.

Nevertheless, the reservoir and well parameters obtained from the analysis were of the same order as those obtained using the MDT tool ($k=100$ and $S=50$). However, the total layer testing time using the new formation tester was about 1 h compared with 20 h (swabbing plus TST test) to get the same information.

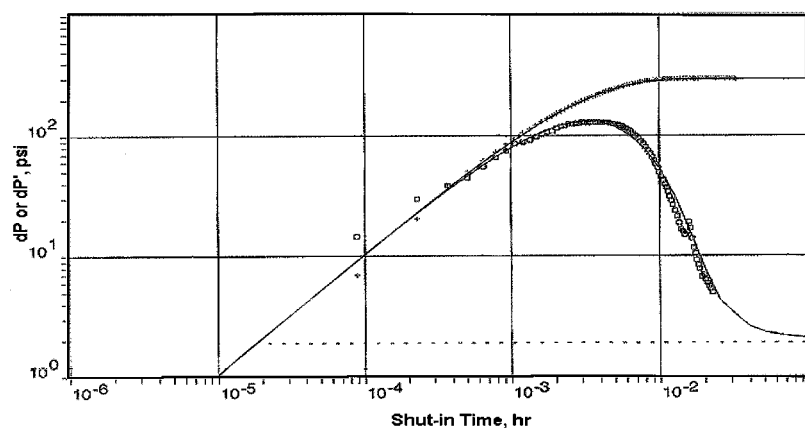


Fig. 6—Log-log plot of the pressure and pressure derivative data obtained during field case 1 from buildup period before sampling.

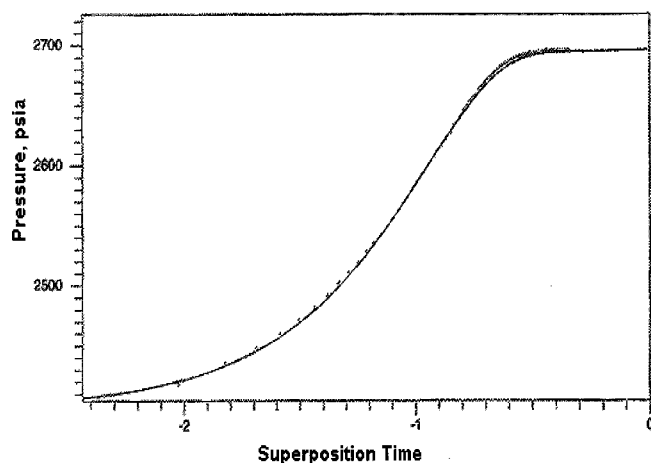


Fig. 7—Superposition plot (field case 1).

The skin value (67) was used as the basis to decide to fracture treat the zone. A nodal analysis (see Fig. 11) was performed using the permeability and damage values obtained from the MDT tool. Surface production predictions were confirmed by actual rates before and after the fracture treatment.

Field Case 2. As noted in the “Introduction,” many of the layers are laminated. Although laminations can be identified using induction logging tools designed for high vertical resolution, quantifying the effect of laminations, for instance, by knowing the value of anisotropy for the layer, is important to understand the production mechanism.

Perforation strategy can be optimized by knowing the value of anisotropy. In this case, besides conducting a layer potential evaluation using the methodology explained in field case 1, a vertical interference test was conducted to obtain vertical transmissibility. Of the four layers tested, the results from one layer are presented here.

The dual probe module of the MDT tool was positioned at X027.83 m. Fig. 12 shows the induction logs for that interval. Gas was identified for the layer, confirming logging predictions. Fig. 13 shows the testing results in graphical form. The induced pulse in the layer is well defined, with a maximum intensity of about 180 cm³/s.

The effect of the pulse was not felt by the vertical pressure monitor located 70 cm vertically above the sink probe that generated the pulse. The high-resolution pressure sensors of the MDT tool can detect pressure changes below 1 psi. A value of k_v/k_h

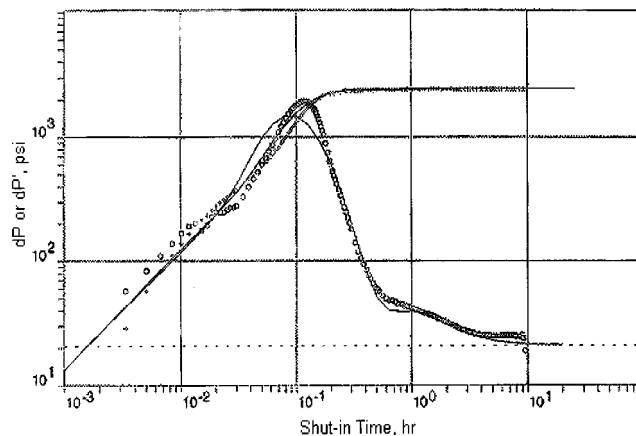


Fig. 9—Diagnostic plot for pressure buildup data obtained from TST in cased hole (field case 1). A homogeneous reservoir model was applied to match the real data.

$=0.06$ was obtained from the analysis. As seen in Fig. 13, a considerable pressure change was recorded by the horizontal monitor (about 17 ps maximum) but almost no pressure change was observed at the vertical monitor. One explanation may be that the high horizontal permeability limited vertical transmission of the perturbation. For this case, the obtained value of anisotropy should be considered qualitative. If the layer is retested to quantify the value of anisotropy, the test design should consider different pulse amplitudes [Ref. 3].

Field Case 3. One of the challenges in the exploratory stage in San Jorge Gulf basin reservoirs is the identification of reserves associated with layers not present in producing wells (i.e., the discontinuous nature of the basin's reservoirs).

Early (openhole) identification of the potential associated with these layers may avoid the cost of setting hundreds of meters of casing just to conduct an evaluation. In this example, three layers were evaluated using the same approach of the previous two cases. The results obtained for one layer are presented here. On the basis of induction logs, the packer module was set at a depth of X22.0 m. In addition to the induction logs, nuclear magnetic resonance (NMR) logs also were run (see Fig. 14). The optical fluid analyzer (OFA) identified oil as the predominant fluid associated with this layer. In addition to fluid identification, another objective was to compare permeability inferred using the NMR log for the layer and the permeability obtained using the MDT packer module.

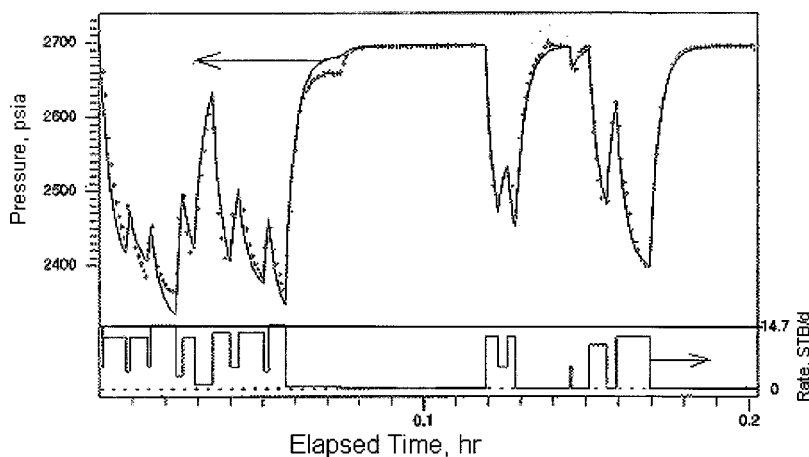


Fig. 8—Layer testing validation of pressure history (field case 1).

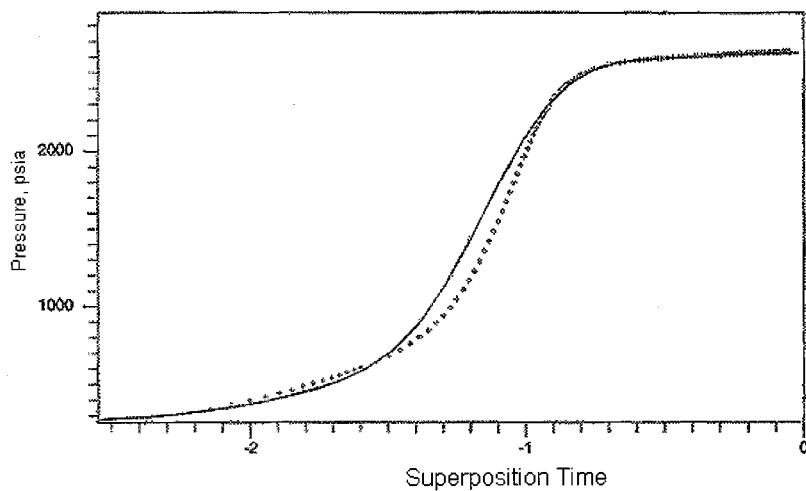


Fig. 10—Superposition plot applied to pressure buildup data obtained from TST in cased hole (field case 1).

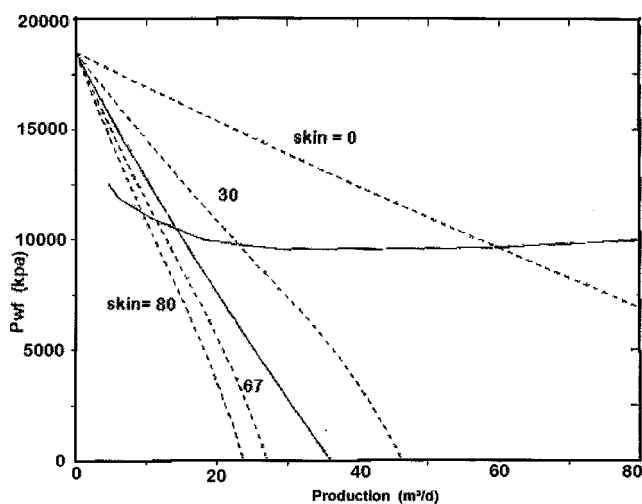


Fig. 11—System analysis for the example presented in field case 1. Production rates before and after the fracture treatment were as expected.

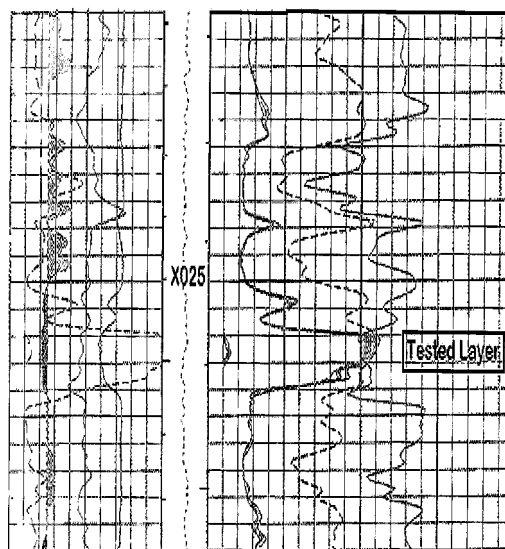


Fig. 12—Induction log showing resistivity response (field case 1) for one layer subjected to a vertical interference test.

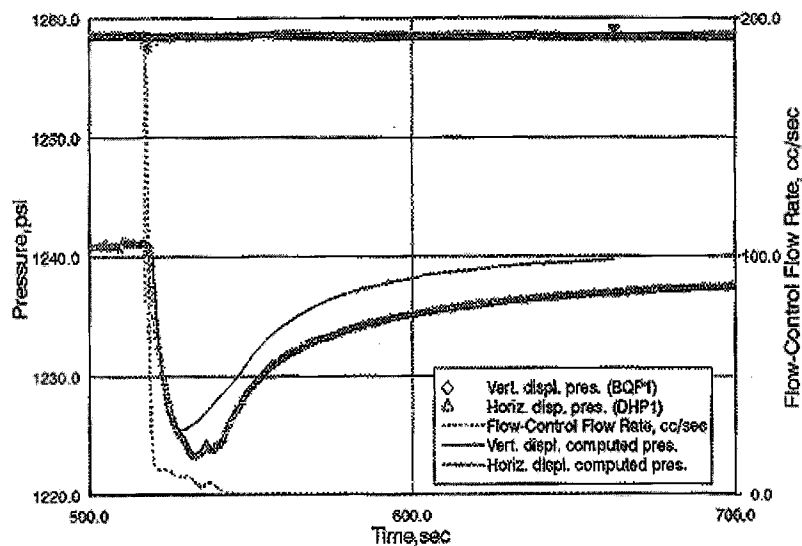


Fig. 13—Vertical interference test results (field case 1). The horizontal pressure monitor responded almost immediately to the operated pulse.

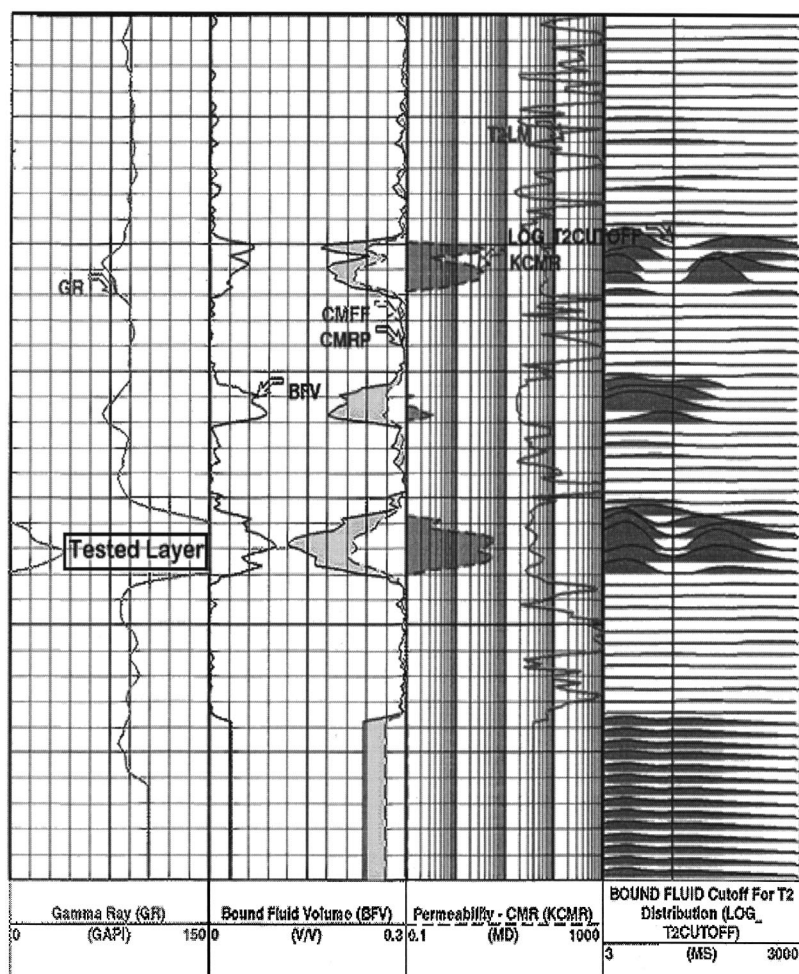


Fig. 14—Nuclear magnetic resonance log from field case 3. The average log-derived permeability for the layer is 5 md.

The sequence of events for the layer testing is shown in Fig. 15. A diagnostic plot of the pressure and pressure derivative for the last buildup is shown in Fig. 16. The pressure transient behavior shows changing wellbore storage effects at early times [Ref. 2].

It is striking how the model can reproduce the real pressure response given that the flow-rate history is characterized by intermittent periods of flow and shut-in. Rates of 4 m³/d are indicative of the potential of the zone. A permeability of 6.9 md and a skin factor of -1.1 were obtained from the analysis of pressure transient data. The layer pressure is 1,161.07 psia.

According to the NMR log, the layer has an average log-derived permeability of 5 md.

Field Case 4. As mentioned in the introduction section the new approach of layer evaluation is not limited to the San Jorge Gulf basin only. This field case deals with the application of the methodology in wells located in the Neuquen Basin. The Chihuido de La Sierra Negra oil field is the most representative field of this basin. The general characteristics of the producing layers of this field can be found in Ref. 4. Layer's thickness are in general thicker than the San Jorge Gulf basin. Layer fluid identification is not a challenge in this case, however, there are cases where the setting of hundreds of meters of casing depends on accurate information of the layer fluid (water or oil). The layer logging and testing procedures are similar to the one used in the San Jorge Gulf basin. The test objectives in this case were to compare the obtained results for permeability, damage, and layer pressure using the new approach with the values obtained using conventional TSTs after zone completion. Based on the nuclear magnetic reso-

nance log which defines the interval for testing (about 15 m thick), the dual packer module was set at a depth of X059.9 m. Figs. 17 and 18 show diagnostic plots corresponding to the buildup period and test validation as a quick check on the values of permeability and damage ($k=2.55$ md and $S=0.48$). The estimated layer pressure was 341 psia. Based on the diagnostic plot given by the pressure and pressure derivative, true radial flow is reached at

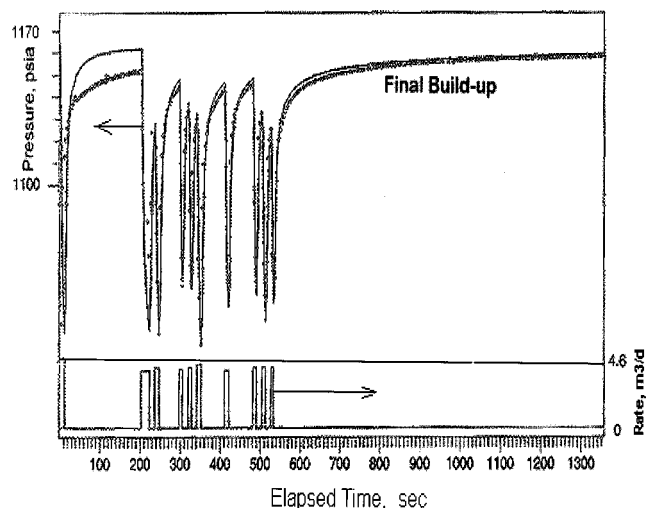


Fig. 15—Layer testing sequence for field case 3. A homogeneous reservoir model was applied to match the data.

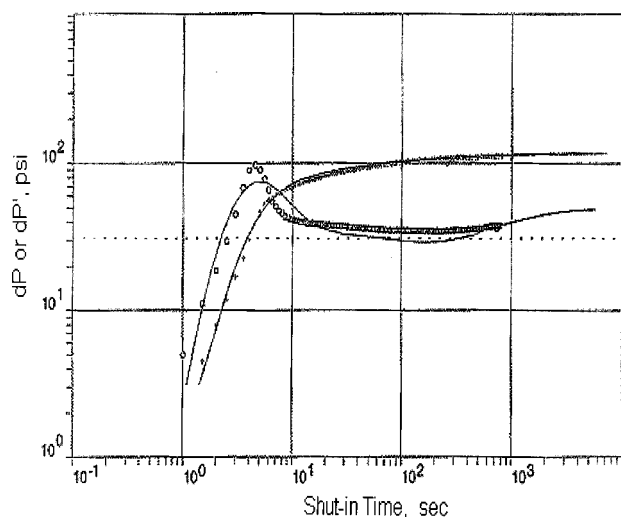


Fig. 16—Log-log plot of the last pressure buildup data (field case 3). A homogeneous reservoir model was applied to match the real data.

about 0.008 h (29 sec). The total test duration was about 0.78 h. The well was cased, and a TST was conducted in the same interval. Conducting this test requires swabbing the well for about 8 to 9 h (no pressure measurement were taking during this period) and finally shutting in the well. Similar testing procedure was conducted as explained in field case 1. The flowing pressure is obtained by letting fluid enter the well (increasing the liquid level) and measuring the pressure during the period (about 0.8 h) preceding the buildup (9.31 h). **Figs. 19 and 20** show diagnostic plots of the pressure and pressure derivative for the buildup period and test validation. The diagnostic plots of the pressure and pressure derivative show that almost all the buildup period is affected by changing wellbore storage effects. Radial flow regime is not well defined. The estimated values for permeability and damage were: $k=1.47$ md, $S=-3$, and the estimated formation pressure was 386 ps. Compared to the results obtained using the dual packer module, even though they are consistent with the exemption of the skin factor, there is no quality control on the TST data. In particular, the flow history prior to the buildup period needed for a proper analysis is not known (swabbing period).

Table 1 presents a summary of the main results obtained in two of the field cases (field cases 1 and 4).

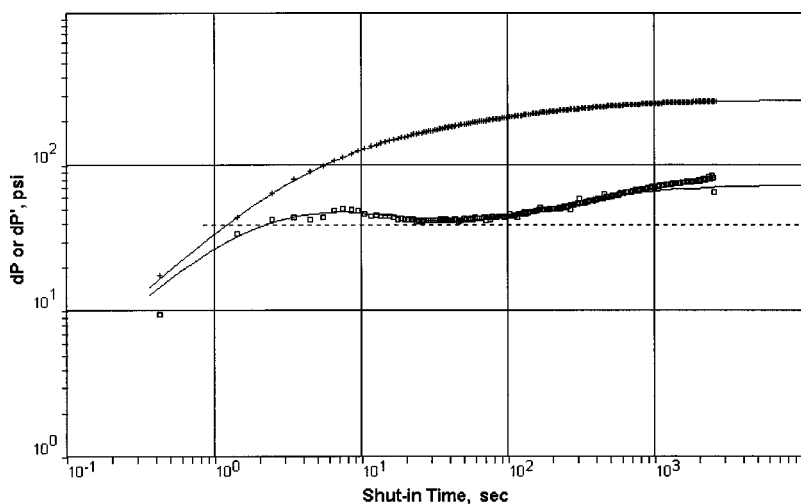


Fig. 17—Log-log plot of the pressure buildup data (field case 4). A homogeneous reservoir model was applied to match the real data.

Layer Testing: Data Analysis Evaluation

In this approach, in addition to fluid identification accomplished in real time using the OFA module, an important issue is the determination of reservoir parameters such as permeability and formation damage (skin). These parameters are obtained from pressure transient analysis of the drawdown, buildup, and interference data. Conventional well test reservoir models were applied to analyze the data. Although the data validation can be considered successful and striking in many cases (see Fig. 8), information on geology and petrophysics is needed to obtain a good match of all acquired pressure data with conventional models. Layer heterogeneity, as well as changing wellbore storage effects in some cases, induces a transient pressure behavior that cannot be reproduced easily using present reservoir models (see Fig. 16). This should be considered characteristic of layer fluid flow behavior for this basin. Testing time management is needed at the wellsite to retrieve good quality pressure data. To get a well-defined radial flow period during buildup, the duration should be 4 to 5 times longer than that of the preceding drawdown period. Nevertheless, measurement in real time remains a significant advantage for better defining testing time to obtain radial flow. To summarize, there are limitations that should be considered when using the MDT method. In particular, to identify virgin formation fluid in the case of a low transmissibility layer and with a high degree of filtrate invasion, it may require testing and pumping time in excess of 1 h to retrieve virgin formation fluid. Because we are at openhole conditions, there is a limit of 8 h for the total testing time/layer. This is a safety rule associated with the packer elements.

Conclusions

Based on these field cases (and many others not presented here), it may be concluded that openhole evaluation of layer potential using the MDT tool is appropriate for San Jorge Gulf and Neuquen basin reservoirs. Many of the layers are 2 to 15 m thick. When testing is conducted with the dual packer module (1 m of a layer is isolated for testing) the whole formation is exposed for well testing. The lack of partial penetration effects observed in many of the tested layers reflects the highly laminated nature of the layers.

Although this is a simple technique from an operations point of view, testing time management at the well site is required to obtain reservoir parameters such as permeability and damage from pressure transient analysis of the data.

Economic evaluation of the method is not addressed here. However, application of this method depends on the associated

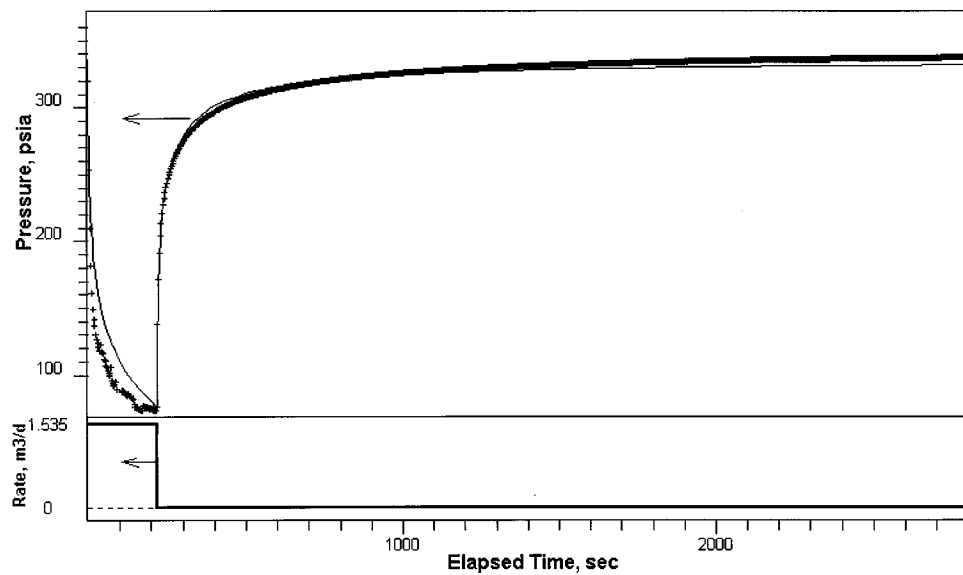


Fig. 18—Layer validation sequence for field case 4. Homogeneous reservoir model considering the presence of a sealing fault was applied to match the data.

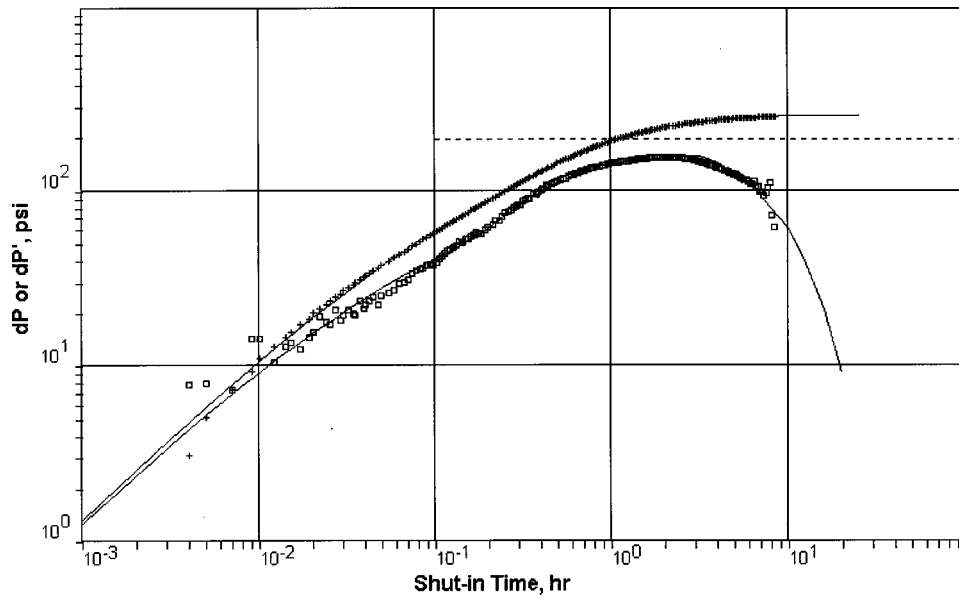


Fig. 19—Diagnostic plot for pressure buildup data obtained from TST in cased hole (field case 4). A homogeneous reservoir model was applied to match the real data.

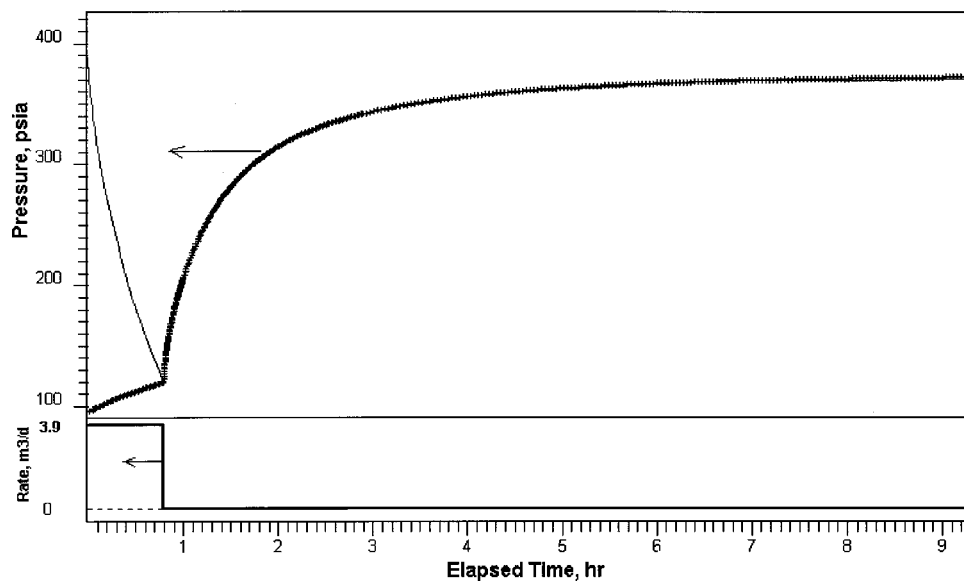


Fig. 20—Layer testing sequence for field case 4 (TST cased-hole data). A homogeneous reservoir model was applied to match the real data.

TABLE 1—COMPARISON OF MDT AND TST TESTING METHODS

Reservoir Parameters	Field Case 1		Field Case 4	
	MDT	TST	MDT	TST
fluid type	oil	oil	oil	oil
permeability (md)	2.55	1.47	116	100
pressure (psia)	340	386	2697	2632
skin	0.5	−3	67	50
testing time (h)	0.78	19	1	20

economic benefits, for example, identifying reserves, saving drilling rig time, avoiding completion of a zone and identifying candidate layers for fracture treatments, etc. A statistical study of the treatments performed is under way to quantify the economic benefits.

Based on the work presented here, the following conclusions are made.

1. Fluid type identification based on logging techniques can be confirmed in real time using the new-generation MDT tool for wireline testing. The layer is not completed if fluid is not of economic interest at the present time.

2. Individual layer testing can be conducted in real time and at openhole conditions in the multilayer environment that characterizes San Jorge Gulf basin wells or any other multilayer environment. Total test time of one hour or less is significantly less than conventional analysis. Layer pressure, reservoir permeability, and formation damage are calculated at the wellsite.

3. The layer anisotropy (ratio of vertical to horizontal permeability) obtained is valid on a layer scale. Conductive laminations can be inferred to define the perforation strategy.

4. Based on the field cases, the values of permeability and damage calculated by layer testing at openhole conditions are in agreement with cased hole DST results.

5. Early (openhole) evaluation of individual layer potential should always be conducted to apply optimal completion procedure.

Nomenclature

- dP = pressure change, psi
 dP' = derivative of pressure change, psi
 k = permeability, L^2 , md
 P_{wf} = wellbore flowing pressure, kPa
 S = skin factor, dimensionless

Subscripts

- h = horizontal
 v = vertical

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SI Metric Conversion Factors

- $\text{ft} \times 3.048 = \text{m}$
 $\text{md} \times 9.869\,233 = \mu\text{m}^2$
 $\text{psi} \times 6.894\,757 = \text{kPa}$
 $\text{barrel/day} \times 1.589\,873 = \text{m}^3/\text{day}$

SPERE

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